FUN-3.C.1 The chain rule provides a way to differentiate composite functions.

3-1 The Chain Rule

Get Ready...

Composite Functions are functions that are of the form $f(g(x))$. They consist of an "inner" function $g(x)$, and an "outer" function $f(x)$. Fill in the chart below for the following functions.

Function	Is the function a composite function? $f(g(x))$ (yes/no)	If yes, what is $\boldsymbol{g(x)}$, the **"inner"** function	If yes, what is $\mathbf{f(x)}$, the **"outer"** function
$y = x^3 + 5x$	No	------	------
$y = (x^2 + 4x)^5$	Yes	$x^2 + 4x$	x^5
$y = \cos(8x^3)$	Yes	$8x^3$	$\cos(x)$
$y = x^4 \sin x$	No	------	------
$y = e^{5x}$	Yes	$5x$	e^x
$y = 8xe^x$	No	------	------

To differentiate these composite functions, we need a new rule:

The Chain Rule	Given g is differentiable at x and f is differentiable at $g(x)$, and $F(x) = f\big(g(x)\big)$, then $$\frac{d}{dx}\big[f\big(g(x)\big)\big] = f'\big(g(x)\big) \cdot g'(x)$$ In words, "the derivative of the outer function, evaluated at the inner function, times the derivative of the inner function." We still only take 1 derivative at a time! We take the OUTSIDE derivative first—leaving the inside the same—then MULTIPLY by the derivative of the INSIDE.

Note: We only use the chain rule with composite functions. Look for "stuff inside parentheses!"

We can combine all our derivative rules with the chain rule. To make it easier to write, let the "inner" function $g(x) = u$.

Derivative Rules Redux (Including the Chain Rule)	
Function $f(x)$	**Derivative $f'(x)$**
$f(x) = u^n$	$f'(x) = nu^{n-1} \cdot \frac{du}{dx}$
$f(x) = \sin(u)$	$f'(x) = \cos(u) \cdot \frac{du}{dx}$
$f(x) = \cos(u)$	$f'(x) = -\sin(u) \cdot \frac{du}{dx}$
$f(x) = \tan(u)$	$f'(x) = \sec^2(u) \cdot \frac{du}{dx}$
$f(x) = \cot(u)$	$f'(x) = -\csc^2(u) \cdot \frac{du}{dx}$
$f(x) = \sec(u)$	$f'(x) = \sec(u)\tan(u) \cdot \frac{du}{dx}$
$f(x) = \csc(u)$	$f'(x) = -\csc(u)\cot(u) \cdot \frac{du}{dx}$
$f(x) = \ln(u)$	$f'(x) = \frac{1}{u} \cdot \frac{du}{dx}$
$f(x) = e^u$ With exponentials, the exponent is ALWAYS u!	$f'(x) = e^u \cdot \frac{du}{dx}$

Example 1: Let $g(x) = \frac{-7}{(2x-3)^2}$. Find $g'(x)$.

$$g(x) = -7(2x-3)^{-2}$$

$$g'(x) = 14(2x-3)^{-3} \cdot (2)$$

$$= \frac{28}{(2x-3)^3}$$

Example 2: Let $f(x) = (3x - 2x^2)^3$. Find $f'(x)$.

$$f'(x) = 3(3x-2x^2)^2(3-4x)$$

or

$$3(3-4x)(3x-2x^2)^2$$

Example 3: Let $g(x) = \sin(4x - 1)$. Find $g'(x)$.

$$g'(x) = \cos(4x-1)\,(4)$$
$$= 4\cos(4x-1)$$

Example 4: Find $f'(x)$ given $f(x) = e^{2-5x}$.

$$f'(x) = e^{2-5x}(-5)$$
$$= -5e^{2-5x}$$

Example 5: Find $f'(x)$ given $f(x) = \frac{4}{\sqrt{x^2+3}}$. $= \frac{4}{(x^2+3)^{1/2}} = 4(x^2+3)^{-1/2}$

$$f'(x) = -2(x^2+3)^{-3/2}(2x)$$

$$= -4x(x^2+3)^{-3/2}$$

OR

$$\frac{-4x}{(x^2+3)^{3/2}}$$

$$\cancel{\sqrt{(x^2+3)^3}}$$

Example 6: Find $f'(x)$ given $f(x) = \sin^2(3x) = (\sin(3x))^2$

$$f'(x) = 2(\sin(3x))\cos(3x) \cdot 3$$
$$= 6\sin(3x)\cos(3x)$$

Example 7: Find $g'(x)$ given $g(x) = \ln(4x^2 - 5x + 7)$.

$$g'(x) = \frac{1}{4x^2-5x+7} \cdot (8x-5)$$

$$= \frac{8x-5}{4x^2-5x+7}$$

Example 8: Let $h(x) = f(x^3)$. Find $h'(x)$.

$$h'(x) = f'(x^3) \cdot 3x^2$$

Example 9: Let $k(x) = g(f(x))$. Find $k'(x)$.

$k'(x) = g'(f(x)) \cdot f'(x)$

x	-3	-1	0	1	2	3	8
$f(x)$	1	0	7	-9	-1	2	4
$g(x)$	6	2	3	0	-3	-4	2
$f'(x)$	4	7	11	-7	6	7	3
$g'(x)$	0	3	-6	4	-5	-1	5

Example 10: Let $h(x) = f(3x^2)$. Find $h'(-1)$.

$h'(x) = f'(3x^2) \cdot (6x)$

$h'(-1) = f'(3)(-6)$

$= 7(-6) = -42$

x	-3	-1	0	1	2	3	8
$f(x)$	1	0	7	-9	-1	2	4
$g(x)$	6	2	3	0	-3	-4	2
$f'(x)$	4	7	11	-7	6	7	3
$g'(x)$	0	3	-6	4	-5	-1	5

Example 11: Let $k(x) = f(g(x))$. Find $k'(2)$.

$$k'(x) = f'(g(x)) \cdot g'(x)$$

$$k'(2) = f'(g(2)) \cdot g'(2)$$

$$= f'(-3)\, g'(2)$$

$$= 4(-5) = -20$$

x	-3	-1	0	1	2	3	8
$f(x)$	1	0	7	-9	-1	2	4
$g(x)$	6	2	3	0	-3	-4	2
$f'(x)$	4	7	11	-7	6	7	3
$g'(x)$	0	3	-6	4	-5	-1	5

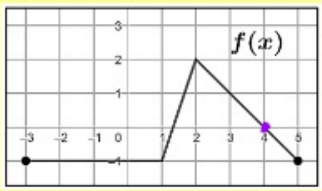
Example 12: Let $n(x) = g(2x - x^2)$. Find $n'(2)$.

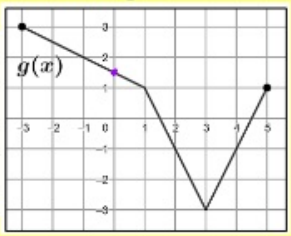
$n'(x) = g'(2x-x^2)(2-2x)$

$n'(2) = g'(0)(-2)$

$= -6(-2) = 12$

The graphs of $f(x)$ and $g(x)$ are above and each consist of 3 line segments. Use the graphs to find the following.





Example 13: Let $h(x) = g\big(f(x)\big)$. Find $h'(4)$.

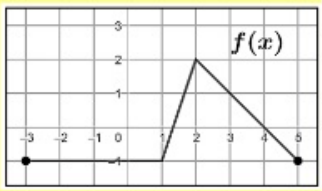
$$h'(x) = g'(f(x)) \cdot f'(x)$$

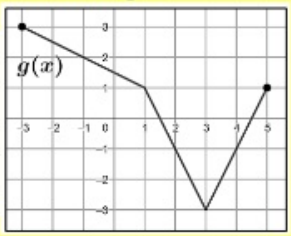
$$h'(4) = g'(f(4)) \cdot f'(4)$$

$$= g'(0) \cdot f'(4)$$

$$= \frac{-1}{2}(-1) = \frac{1}{2}$$

The graphs of $f(x)$ and $g(x)$ are above and each consist of 3 line segments. Use the graphs to find the following.





Example 14: Let $p(x) = g(\ln(x))$. Find $p'(1)$.

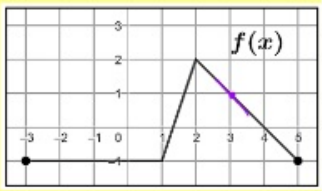
$$p'(x) = g'(\ln x) \cdot \frac{1}{x}$$

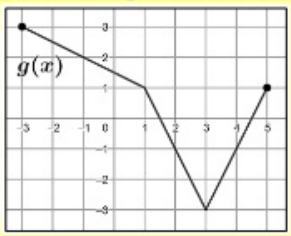
$$p'(1) = g'(\ln 1) \cdot \frac{1}{1}$$

$$= g'(0)$$

$$= -\frac{1}{2}$$

The graphs of $f(x)$ and $g(x)$ are above and each consist of 3 line segments. Use the graphs to find the following.





Example 15: Let $L(x) = e^{f(x)}$. Find $L'(3)$.

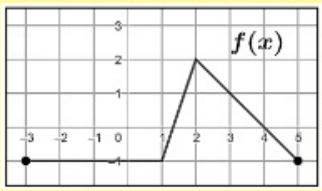
$$L'(x) = e^{f(x)} \cdot f'(x)$$

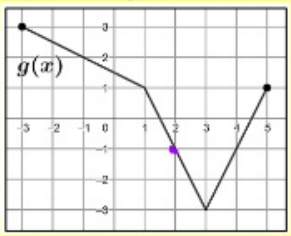
$$L'(3) = e^{f(3)} \cdot f'(3)$$

$$= e^{1}(-1)$$

$$= \boxed{-e}$$

The graphs of $f(x)$ and $g(x)$ are above and each consist of 3 line segments. Use the graphs to find the following.

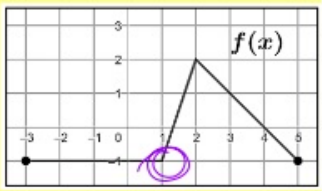


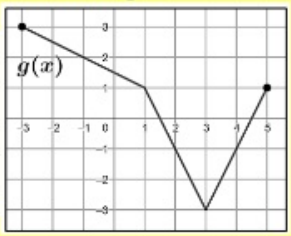


Example 16: Let $B(x) = \sin(g(x))$. Find $B'(2)$.

$$B'(x) = \cos(g(x)) \cdot g'(x)$$
$$B'(2) = \cos(g(2)) \cdot g'(2)$$
$$= \cos(-1) \cdot (-2)$$
$$= -2\cos(-1)$$

The graphs of $f(x)$ and $g(x)$ are above and each consist of 3 line segments. Use the graphs to find the following.

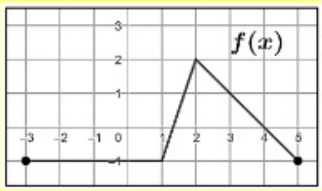


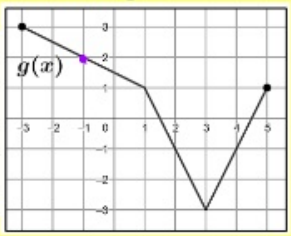


Example 17: Let $q(x) = f(\cos(x))$. Find $q'(0)$.

$$q'(x) = f'(\cos x) \cdot (-\sin x)$$
$$q'(0) = f'(\cos 0) \cdot (-\sin 0)$$
$$= f'(1) \cdot 0$$
$$= \text{undefined}$$

The graphs of $f(x)$ and $g(x)$ are above and each consist of 3 line segments. Use the graphs to find the following.





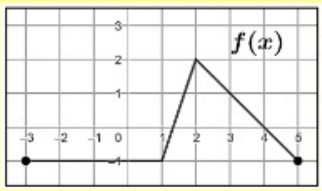
Example 18: Let $r(x) = \ln(g(x))$. Find $r'(-1)$.

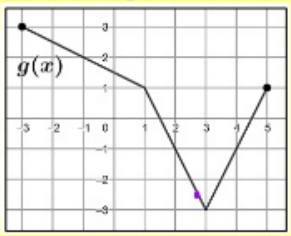
$$r'(x) = \frac{1}{g(x)} \cdot g'(x)$$

$$r'(-1) = \frac{1}{g(-1)} \cdot g'(-1)$$

$$= \frac{1}{2}\left(\frac{-1}{2}\right) = \frac{-1}{4}$$

The graphs of $f(x)$ and $g(x)$ are above and each consist of 3 line segments. Use the graphs to find the following.





Example 19: Let $D(x) = g(e^x)$. Find $D'(1)$.

$e = 2.71828\ldots$

$D'(x) = g'(e^x) \cdot e^x$

$D'(1) = g'(e^1) \cdot e^1$

$= -2e$

Example 20: Find $g'(x)$ given $g(x) = (5x+2)^3(3x+2)^5$.

$$g'(x) = 3(5x+2)^2(5)(3x+2)^5 + (5x+2)^3(5)(3x+2)^4(3)$$

$$= 15(5x+2)^2(3x+2)^4[3x+2+5x+2]$$

$$= 15(5x+2)^2(3x+2)^4(8x+4)$$

$$= 60(5x+2)^2(3x+2)^4(2x+1)$$

Example 21: Find $f'(t)$ given $f(t) = \left(\frac{t-2}{2t+1}\right)^9$.

$$f'(t) = 9\left(\frac{t-2}{2t+1}\right)^8 \cdot \frac{5}{(2t+1)^2}$$

$$= \frac{9}{1}\,\frac{(t-2)^8}{(2t+1)^8} \cdot \frac{5}{(2t+1)^2}$$

$$= \frac{45(t-2)^8}{(2t+1)^{10}}$$

$$\frac{(2t+1)1-(t-2)(2)}{(2t+1)^2}$$

$$\frac{2t+1-(2t+4)}{(2t+1)^2}$$

$$\frac{5}{(2t+1)^2}$$

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or use the quotient rule

$$f(t) = \frac{(t-2)^9}{(2t+1)^9}$$

$$f'(t) = \frac{(2t+1)^9(9)(t-2)^8 - (t-2)^9(9)(2t+1)^8(2)}{(2t+1)^{18}}$$

$$= \frac{9(2t+1)^8(t-2)^8\left[2t+1-2(t-2)\right]}{(2t+1)^{18}}$$

$$\frac{9(2t+1)^8(t-2)^8(5)}{(2t+1)^{18}} = \frac{45(t-2)^8}{(2t+1)^{10}}$$